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RESEARCH

Fungal Communities Shift with Soybean Cyst Nematode Abundance in Soils

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ABSTRACT

Plant parasitic nematodes pose a significant threat to food security. Management strategies for these pathogens are limited, and additional tools for their control are needed. Some fungi have shown promise for biocontrol; however, their success in the field has varied. In contrast, some fungal plant pathogens form synergistic associations with nematodes, resulting in increased plant disease severity. However, how the two groups of fungi change with different plant parasitic nematode abundances in the soil is underexplored. In this study, we used the soybean cyst nematode as a model to understand these changes. We sampled soil from 171 Ohio soybean fields in 2019 and 2021 and determined soybean cyst nematode abundance. We identified the fungi in the samples through amplicon sequencing of the 18S-ITS rDNA regions. Edaphoclimatic factors were used to classify samples into geographic regions to account for environmental differences between sampling locations. We hypothesized that fungal communities would be influenced by both region and soybean cyst nematode abundance. K-near-neighbor analysis revealed that fungal communities follow regional patterns. We also found that soybean cyst nematode abundance was associated with changes in these communities regardless of the region. Two potential nematophagous fungi were found to be prevalent in Ohio through core community analysis, although they were enriched when soybean cyst nematode abundance was high. Finally, differential network analysis showed that interactions among fungal community members change when soybean cyst nematode is present in the soil. Together these results suggest that this nematode significantly shifts the fungal community composition in field soils.

Keywords: mycobiome, plant parasitic nematodes, soil fungi, soybean, soybean cyst nematode

Plant parasitic nematodes (PPN) pose a significant threat to global food security, with estimated losses of more than $100 billion every year (Elling 2013). Managing PPN can be challenging because the use of pesticides against these microorganisms is strictly regulated because of toxicity. Crop rotation and the use of resistant cultivars are the primary strategies employed for nematode control (Winsor 2020). However, the repeated use of the same sources of resistance leads to a change in virulence among nematode populations. This enables PPN to reproduce on resistant cultivars, restricting management options (Djian-Caporalino et al. 2011; Mitchum 2016). Therefore, new strategies that can effectively reduce PPN damage are needed. The use of microorganisms for biological control has been explored, and some fungi have shown promising parasitic activity against PPN (Haarith et al. 2020, 2021; Kim and Riggs 1991, 1995; Timper and Riggs 1998). However, successful biocontrol application in the field has been challenging (Godoy et al. 1982; Haarith et al. 2020; Ma et al. 2005; Meyer et al. 1990). Many factors contribute to this, including our limited understanding of the interactions between native soil microbiota and populations of PPN, which could influence biocontrol efficacy. Several studies have demonstrated that effective suppression of PPN can be achieved through increased abundance of native microbial nematode antagonists (Khan et al. 2021; Smith Becker et al. 2024; Topalović et al. 2020). To identify factors that contribute to the success of biocontrol agents in the field, it is important to recognize that antagonistic interactions do not occur in isolation.

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They are influenced by other microorganisms and the surrounding environment.

One of the most economically important PPN is *Heterodera glycines*, the soybean cyst nematode (SCN). SCN is the most important yield-limiting pathogen of soybean (*Glycines max*) in North America (Bandara et al. 2020). In the United States, which ranks second in soybean production worldwide, SCN causes losses exceeding $16,000 per planted hectare (Bandara et al. 2020; USDA 2023). SCN is an endoparasite that travels through the soil until it finds the roots of a suitable host. Once inside the root, the nematode initiates feeding sites, becomes sedentary, and develops into an adult. Fertilized female nematodes lay eggs and eventually become filled with them. After its death, the swollen female's body hardens, forming a durable cyst that protects the eggs for many growing seasons (Inagaki and Tsutsumi 1971). This makes management of SCN challenging owing in part to its persistence once it infests a field. The use of resistant cultivars has been vital in managing SCN damage in soybean fields. Unfortunately, SCN populations are overcoming resistance in many commercially available cultivars (McCarville et al. 2017; Niblack et al. 2008). Given the significant role of soybean worldwide, it is imperative to mitigate the impacts of pathogens like SCN (Hartman et al. 2011). SCN alone can be yield limiting, but SCN can also interact synergistically with fungal soybean pathogens, resulting in additional crop losses (Lopez-Nicora et al. 2016, 2020; Roth et al. 2019; Xing and Westphal 2013). On the other hand, several groups of fungi have been identified as antagonists to SCN and other PPN (Haarith et al. 2020; Kim and Riggs 1991, 1995; Timper and Riggs 1998; Wei et al. 2009). Although previous studies have explored some of these interactions in the rhizosphere and have described the nematode mycobiome, little is known about how fungal communities change with PPN abundance in soils (Elhady et al. 2018; Gu et al. 2022; Hu et al. 2018; Jiang et al. 2022; Klein et al. 2013).

Here we use SCN as a model to better understand the relationships between PPN and soil fungi. We considered how physiography and environmental factors contribute to shaping these communities. Additionally, we evaluated SCN abundance as a factor contributing to changes in fungal communities. We hypothesized that fungal community composition aligns with regional patterns. Additionally, we anticipated that SCN impacts soil fungi, and we expected a higher prevalence of fungi identified as nematode antagonists in fields with lower SCN. To test these hypotheses, we sampled soils from Ohio's soybean production fields. We processed these samples to determine SCN abundance and performed synthetic long-read amplicon sequencing to characterize the fungal communities. We also measured edaphic factors and noted the mean annual temperature and precipitation at each location. We observed that fungal communities in Ohio exhibit regional patterns, but regardless of the region, SCN had an impact on the fungal community composition. Both findings supported our initial hypothesis. Only two potential SCN antagonistic fungi were prevalent in our sampling. They were more abundant in fields with higher SCN when compared with fields with low or no SCN detection, contradicting our hypothesis.

MATERIALS AND METHODS

Survey area and sample collection. Soil samples were collected from soybean production fields in Ohio in the years 2019 and 2021. A total of 171 samples were collected from 26 counties, following the recommended sampling approach for detecting SCN (Dorrance et al. 2018). Briefly, a 2.5-cm-diameter soil probe was used to obtain soil cores from each field. A total of 15 soil cores, collected at a depth of 20 cm, were sampled in a zig-zag pattern across an area spanning up to eight hectares in each field. The collected cores, totaling volumes of approximately 2 liters per sampled production field, were stored at 4°C until processing. Soil samples were homogenized by being passed through a 2-mm metal sieve. A subsample of approximately 25 g was then collected and frozen at −80°C until subsequent DNA extraction. To avoid cross-contamination, the sieves were cleaned using 70% EtOH between samples. Frozen soil subsamples were freeze-dried using a Harvest Right home freeze dryer set to a shelf temperature of 15°C and a drying temperature of −15°C for 16 h. The mean annual temperature (MAT) and mean annual precipitation (MAP) for the sampled locations were obtained through the National Centers for Environmental Information. The climate normals of the weather station closest to each sampling location, using the 30-year normals (NCEI 2021), were selected.

Soil analyses. To determine the abundance of SCN in each sample, 100 cm^3 of the collected soil was elutriated using a semiautomatic elutriator with nested sieves of apertures of 600 microns over 180 microns (Byrd et al. 1976) to separate the nematode cysts. From the bottom sieve, the cysts were collected and ground using a drill press with a rubber stopper attachment to release the eggs contained within. The resulting material from the ground cysts was collected on nested sieves with apertures of 75 microns on top and 25 microns on the bottom. Eggs were stored in a 50-ml tube and stained with acid fuchsin for counting. The tube was filled with water to complete a total volume of 50 ml as a starting dilution before a 5-ml subsample of the homogenized dilution was counted using a stereoscope at 5× magnification. The soil samples collected in 2021 were air dried and sent to Brookside Laboratories, Inc. (New Bremen, OH, U.S.A.). There the following tests were done: Bray II phosphorus, Mehlich III Extractable, total exchange capacity determination by summation, organic carbon measured by loss-on-ignition, and soil texture determined by hydrometer (Ashworth et al. 2001; Bray and Kurtz 1945; Mehlich 1984; Ross and Ketterings 1995; Schulte and Hopkins 1996).

DNA extraction and sequencing. Genomic DNA (gDNA) was extracted from 250-mg freeze-dried soil samples. For soils collected in 2019, gDNA was extracted using the SurePrep Soil DNA Isolation Kit by Fisher Scientific (Pittsburgh, PA, U.S.A.) per manufacturer instructions. In short, a provided bead tube was used for sample disruption along with the MP Biomedicals FastPrep-24 5G system set at 4.0 m/s for 30 s. After this, the lysis enhancer and column for humic acid reduction were used. DNA extractions from 2021 samples were done with the NucleoSpin Soil DNA extraction kit by TakaraBio (San Jose, CA, U.S.A.), as the original kit was discontinued. Similar settings were used for sample disruption. The provided SL1 buffer was used along with the optional enhancer SX. An empty extraction reaction was run for each kit as a control. Three positive controls were included in the 2019 sequencing run. Zymo Community standards D6311 and D3605 and ATCC NGS standard MSA-1010 were used. The D6311 Zymo Community standard was used as positive control in 2021. To confirm the gDNA extraction of all samples, we conducted electrophoresis with a 1% agarose gel. The concentration and quality of the gDNA were assessed through spectrophotometry using a Nanodrop 2000 (Thermo-Scientific, Waltham, MA, U.S.A.). Samples were sent to the Loop Genomics (San Diego, CA, U.S.A.) sequencing facility, where library prep and synthetic long-read sequencing of the 18S and ITS regions of the rDNA was performed (Chung et al. 2020). This sequencing approach tags individual molecules of DNA in a complex sample. When the DNA is amplified, the unique tag from the original strand is randomly distributed among various positions within the new fragments (Jeong et al. 2021). These fragments are subsequently sequenced using short-read sequencing and assembled (Jeong et al. 2021).

Bioinformatic pipeline. Once sequences were obtained from Loop Genomics, the ITS1 and ITS2 regions were extracted using *ITSxpress* (Rivers et al. 2018). The extracted sequences went through quality assessment, error correction, dereplication, and removal of sequencing artifacts using the *dada2* pipeline as described in Callahan et al. (2021). From this, amplicon sequence variants (ASV) were produced and classified using the UNITE database (version 8.3) (Abarenkov et al. 2024). The taxonomy table generated along with the samples' metadata and the reference ASV were used to construct a *phyloseq* object (McMurdie and Holmes 2013). Afterward, additional filtering was performed to retain sequences classified under the Kingdom Fungi. Samples with fewer than 100 reads were removed.

Statistical methods. All statistical analyses were done using R version 4.3.0 (R Core Team 2023). To delimit geographic regions, a principal component analysis (PCA) was done with the 2021 samples for which we had information on edaphic factors, MAT, and MAP. This was performed with the "prcomp" function of the *stats* (R Core Team 2023) package, where centering and scaling were also computed. Results were visualized using the *factoextra* (Kassambara and Mundt 2020) package and were used to assign all samples to one of four geographical regions: North, Central, South, and West. Regional influence on fungal communities was assessed using K-near-neighbor analysis through the *phyloseqGraphTest* (Fukuyama 2020) package's "graph_perm_test" function with Jaccard distance measure and a knn value of 2. Crop history effects were analyzed on samples for which data were available using PERMANOVA on Bray-Curtis Distance matrices of community dissimilarity. To identify fungal indicator species of different SCN infestation levels, indicator species analysis was performed. For this, the ASV abundance table was compositionally transformed. Afterward, the "multipatt" function of the *indicspecies* (Cáceres and Legendre 2009) package was used to run the analysis. The group association function "r.g" was selected to account for the uneven number of samples per group and 9,999 permutations were executed. To assess the extent of variation explained by the indicator species, a PCA was conducted using the *microViz* (Barnett et al. 2021) package. The "core_members" function of the *microbiome* (Lahti et al. 2017) package was used to identify core fungal community members at different SCN abundance levels. The analysis was performed with a prevalence setting of 0.75 and minimum detection of 0.001 using the compositionally transformed ASV table. To identify differentially abundant fungi at different levels of SCN abundance, contrast analyses were run using the *DESeq2* (Love et al. 2014) package. Differential network analysis was performed to evaluate microbial network differences in the presence and absence of SCN using the *NetCoMi* (Peschel et al. 2021) package. Samples were grouped by whether SCN was detected in the soil or not before the analysis was performed. To construct the network, the "netConstruct" function was used and the 250 ASV with the highest variability were selected and included in the network. Pseudo counts and centered log-ratio transformation were used to protect against spurious correlations, while Pearson's correlation was used as the association measure with a threshold of 0.4 (Kaul et al. 2017; Kumar et al. 2019). To assess differences in the networks, the "diffnet" function was used with Fisher's z-test as the method to determine differential associations and local false discovery rate as the multiple testing adjustment method. The code used in this study is available at https://github.com/melaniemml/SCN_Fungi_OH.

RESULTS

SCN abundance and distribution. In 2019 and 2021, a total of 171 soil samples were collected from 26 counties in Ohio (Fig. 1). The SCN abundance in the samples ranged from 0 to 15,800 eggs/100 cm^3 of soil. Samples with the highest SCN abundance were predominantly located in the western and northern regions of the state, which are also major soybean production areas. To simplify further analysis and discussion, the SCN abundance was separated into categorical variables (Table 1). Sixty-one samples in which no eggs were found were classified as "Not detected." There were 25 samples classified as having a "Low" SCN abundance, with a range of 1 to 100 eggs/100 cm^3 of soil. Twenty-four samples had a "Medium" SCN abundance (between 99 and 600 eggs/100 cm^3 of soil). And 21 samples with more than 600 eggs/100 cm^3 of soil were classified as "High." These categories were selected with SCN management in mind, where the lower two are not expected to cause significant yield loss for Ohio soybean while the upper two could cause moderate to severe yield losses (Dorrance et al. 2018).

Sequencing summary. A subsample of the soils collected was used for gDNA extraction and 18S and ITS sequencing in two separate runs, 2019 and 2020, respectively. When combined, the runs yielded a total of 719,969 reconstructed raw reads. Reads ranged in length from 10 to 1,878 bp. The average number of reads per sample was 4,210. After filtering and removing samples with poor sequencing depth, the average number of reads per sample rose to 5,057 ($n = 141$). A total of 9,387 ASV were identified with *dada2* (Callahan et al. 2016) and used for downstream analysis. Sequences generated in this study are available in NCBI SRA PRJNA1055501.

Sequencing of negative controls yielded one to four total sequence reads per control, none of which passed the filtering process. The *decontam* (Davis et al. 2018) package, which identifies ASV that could potentially be contaminants, flagged one potential contaminant, which was removed from the data set. Three positive controls were used in the 2019 run, Zymo standards D6311 and D3605 and ATCC standard MSA-1010. These had recovery rates of 100% for the D6311 and 50% for both D3605 and MSA-1010. For the 2020 sequencing run, one positive control was used, Zymo D6311, for which 50% of the fungal constituents were identified. The latter was consistent between the two sequencing runs. Two different DNA extraction kits were used during the different sampling years. Because the year of sample collection can influence the soil's fungal community composition, it is impossible to disentangle the size of the extraction kit's effect on this data set. However, a previous study conducted in our laboratory, using the same sequencing run and extraction kits, revealed that the extraction kits did not significantly contribute to variation in fungal community composition (Frey et al. 2024).

TABLE 1
Summary of the categorical classification of soybean cyst nematode (SCN) abundance based on the number of SCN eggs per 100 cm^3 of soil

Abundance category[a]	SCN eggs per 100 cm^3 of soil	Number of samples[b]
High	>600	21
Medium	100< to ≤600	24
Low	≤100	25
Not detected	0	61

[a] Abundance categories were selected based on SCN impact to soybean yield in Ohio, where the lower two abundance categories are not expected to cause significant yield loss for soybean growers while the upper two categories could cause moderate to severe yield losses (Dorrance et al. 2018).
[b] The number of samples belonging to each abundance category is shown.

Regional features of the fungal community composition. The studied soil samples were collected across the state of Ohio, with known variation in environmental, crop history, and edaphic factors. We recognize that these variations might impact the composition of fungal communities. Therefore, we first evaluated the effect of location on fungal community composition. To achieve this, we conducted a PCA using edaphic factors, as well as MAT and MAP from 2021 samples (samples from 2019 were excluded because of incomplete data). The first five principal components explained 84.8% of the total variation (Supplementary Fig. S1). PC1 and PC2 explained 37.9 and 20.4% of the total variation, respectively (Fig. 2). Such analysis assisted in the delineation of groupings that accounted for both geographical location and environmental factors. Based on the PCA's results (Fig. 2), four regions were identified: Central, North, South, and West, and are highlighted in Figures 1 and 2.

Cropping history (current crop and previous crop) information was available for only a subset of samples (Supplementary Fig. S3), with most sites under soybean as part of a corn and soybean rotation. When analyzed by year, to account for environment and sample processing differences, a significant effect of current crop (i.e., crop at the time of sampling; PERMANOVA, $P < 0.05$) but not of previous year crop (PERMANOVA, $P > 0.05$) was observed on fungal community composition (Supplementary Table S1).

To assess the impact of regional groupings, defined through PCA, on the composition of soil fungal communities, a K-nearest-neighbor analysis was conducted (Fig. 3). We identified the two nearest neighbors for each sample based on their fungal community. We determined whether these neighbors shared a pure edge type, provided that they belonged to the same region. A total of 216 edges were produced out of which 110 were pure and 106 were mixed. Of the pure edge types 54 formed among the Central region samples, 10 in the North region, 8 in the South region, and 38 in the West region. This resulted in 93, 55, 57, and 93% of the samples forming at least one pure edge in each region, respectively.

Soybean cyst nematode influences soil fungal community composition. Indicator species analysis was used to identify the fungi that were significantly ($P \leq 0.05$) associated with different levels of SCN abundance. In total 161 fungi were identified as indicator species, 101 were associated with high SCN abundance, 16 with medium, 39 with low, and one for samples with no SCN detected. Furthermore, two ASV were identified as indicator species for both high and low levels of SCN. One ASV represented the combination of high levels and not detected SCN, while another one the combination of medium levels and undetected SCN (Supplementary Table S2). Using this subset of fungi in a PCA, we evaluated how their community composition changed in our samples as SCN abundance increased (Fig. 4). In this ordination, PC1 explained 20.3% of the total variability, while PC2 accounted for 12.4% of the total variability. The structure in which the samples were ordinated revealed that samples with a high abundance of SCN are more similar to each other compared with samples with other levels of nematode abundance. This similarity is evident

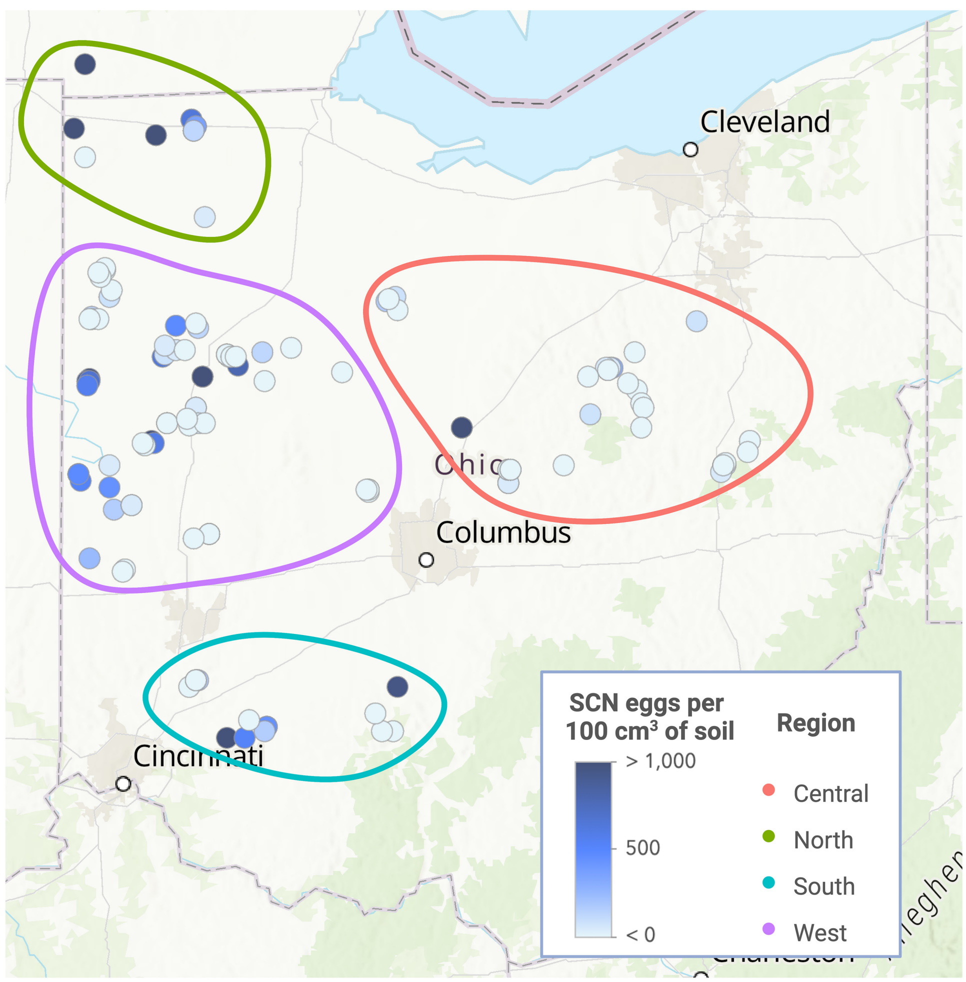


Fig. 1. Map of Ohio with ellipses indicating the locations of sample collection. The color of each circle represents the number of nematode eggs found at that location, with darker shades indicating higher nematode abundance. Different regions are designated by various colors: coral for the central region, green for the northern region, blue for the southern region, and purple for the western region. Regions were defined per the results shown in Figure 2. SCN, soybean cyst nematode.

regardless of the geographical region from which the sample was collected.

A core community analysis was performed to understand which fungi were consistently present in soybean fields throughout Ohio, and at different SCN abundance levels (Fig. 5). In total, six ASV were identified as core members of the fungal community across all samples. Two additional core members were identified in samples with low SCN abundance. In samples with high SCN abundance, four additional core community members were identified. Core community members were also identified for a combination of nematode abundance levels. For the combination of medium nematode abundance and no detection of nematode, there was one additional core community member. This was also true for the combination of low, high, and medium levels of nematode abundance. The low and medium levels of SCN abundance also shared a core community member. Finally, the samples with high and low nematode abundance shared two fungal taxa as core community members. The taxonomic classification for these members is shown in Figure 5.

To further examine the response of soil fungi to SCN abundance, a contrast analysis was conducted. This comparison was made between samples with high SCN abundance and samples where SCN was not detected (Fig. 6). A total of nine ASV were identified as differentially abundant. Among them, six were more prevalent

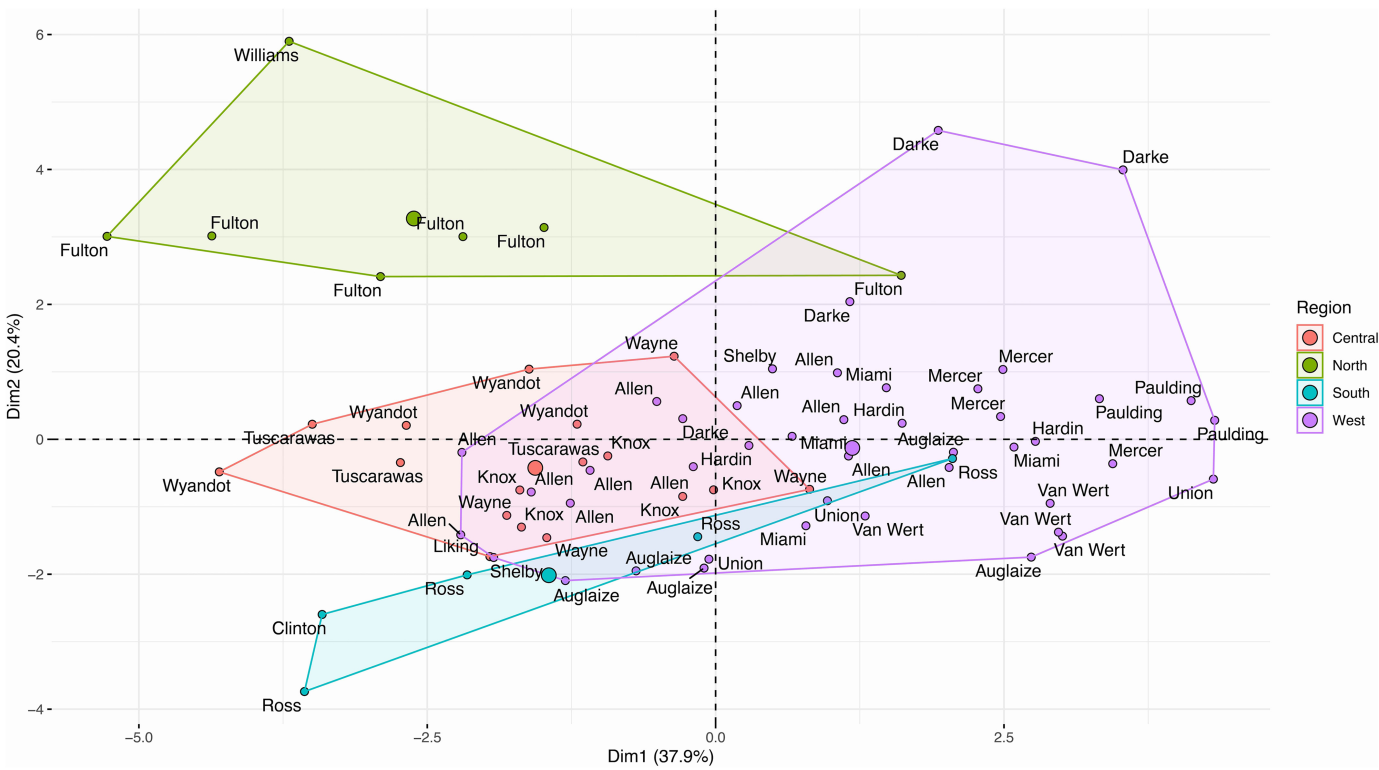
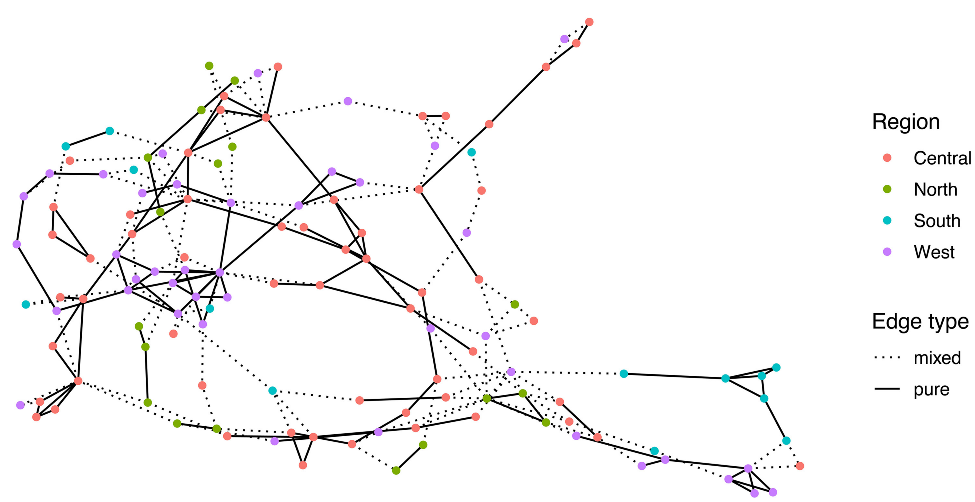


Fig. 2. Principal component analysis of soil samples based on edaphic factors, mean annual temperature, and mean annual precipitation. Each circle represents a sample, with its corresponding county of origin noted. Polygons are drawn to represent different regions, and sample color matches the region of origin. The Central region is depicted in coral color, the North region in green, the South region in blue, and the West region in purple.

Fig. 3. Analysis of fungal community composition in soil samples collected using K-nearest neighbor method. The color of each dot denotes the region from which the sample was collected. The connecting lines between the dots represent the edge type of each near neighbor. Solid lines represent pure edge types, while dotted lines represent mixed edge types (P = 0.002).



when there was no SCN present in the soil: *Tretracladium* sp., *Phallus rugulosus*, *Herpotrichia* sp., *Chaetomium* sp., *Botryotrichum domesticum*, and *B. atrogriseum*. On the other hand, three ASV (*Fusarium esquiseti*, *Chlonostachys rosea*, and *Chaetomium* sp.) were more abundant in samples with high SCN abundance.

Differential network analysis was performed by comparing the association networks of samples with and without SCN detection (Fig. 7; Supplementary Table S3). We found that out of the top 250 ASV with the highest variability between the two groups, 137 formed Pearson's correlations at or above the 0.4 threshold. When comparing these correlations between the two groups, 42 ASV formed significantly different correlations within their respective networks. Among these ASV, those with 95% quantile of all eigenvector centralities in the network were identified as hub nodes within the network. These hub nodes were significantly different between the networks ($P = 0.039$), and the only shared hub node was ASV_64, an unidentified fungus. In the network constructed with SCN-detected samples, four hubs were identified: *Mortierella rishikesha*, *Mortierella* sp., *Linnemannia amoeboidea*, and an unidentified member of the Mortierellaceae family (ASV_74, ASV_96, ASV_115, and ASV_136, respectively). Contrasting the samples where SCN was not detected, the hub taxa in their network were *Ascobolus* sp., *Enterocarpus grenotii*, *Heydenia* sp., and *Mortierella* sp., with the last two as unidentified fungi (ASV_141, ASV_147, ASV_152, ASV_213, ASV_228, and ASV_246, respectively). The core community members *Clonostachys rosea* (ASV_5) and *F. equiseti* (ASV_6) also contributed to the differential network.

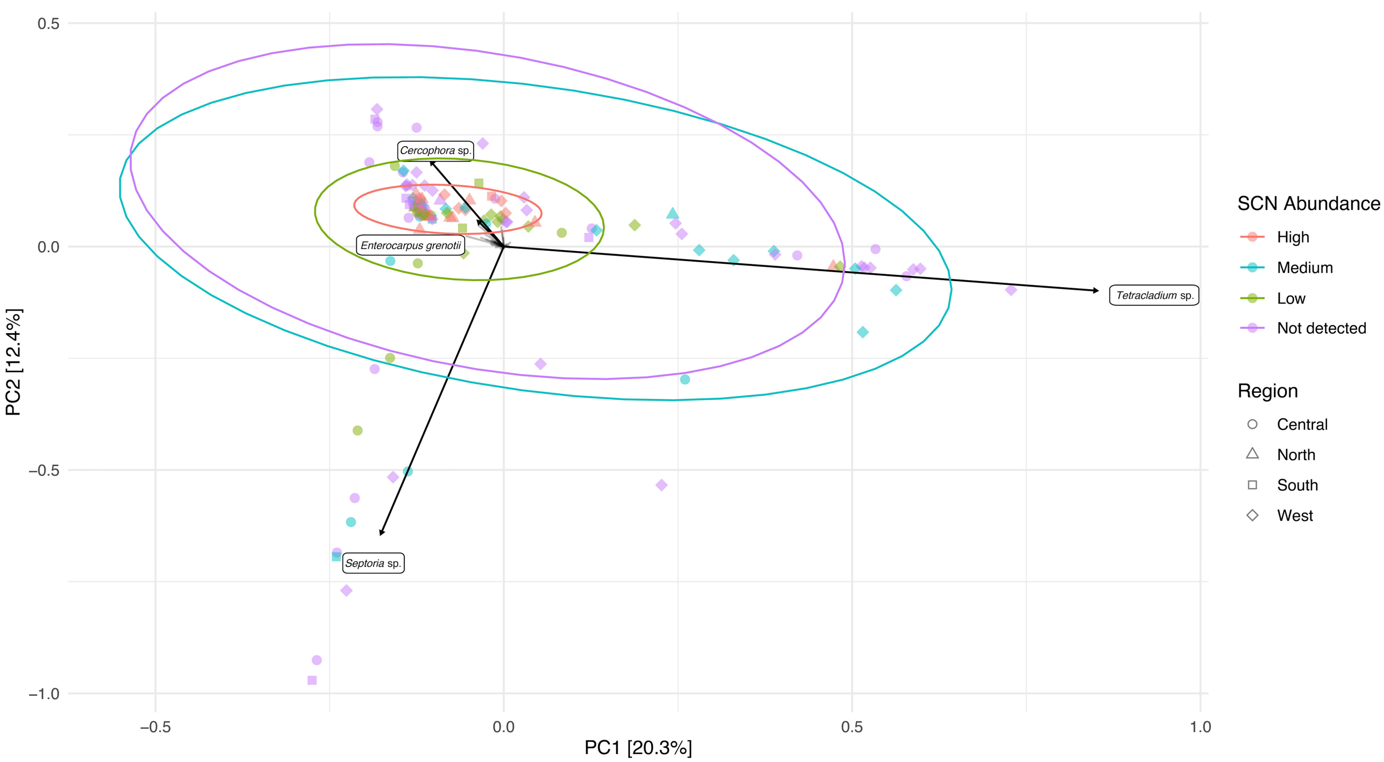


Fig. 4. Principal component (PC) analysis for the fungal community of indicator species in each sample. The ellipse and sample colors represent the classification of soybean cyst nematode abundance in each sample. Ellipses were drawn with a confidence interval of 0.95. Sample shapes indicate the region.

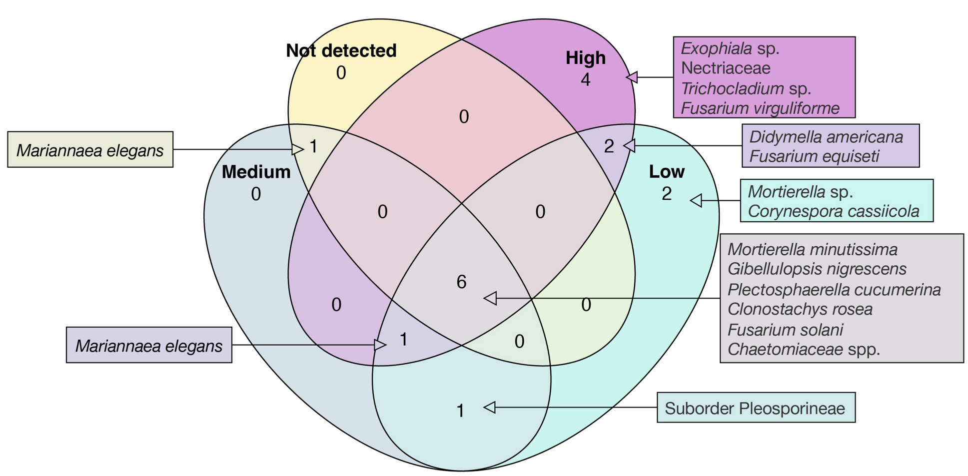


Fig. 5. Venn diagram showing the results of core community composition analysis. Each ellipse contains the core community members at different levels of soybean cyst nematode (SCN) abundance. Unions within the diagram represent shared community members at different levels of SCN abundance. The core microbiome was defined by a minimum detection of 0.001 relative abundance in at least 90% of the samples and a prevalence of 75%.

DISCUSSION

PPN greatly contribute to losses in crop productivity around the world. However, our understanding of how PPN interact with other microorganisms in the soil is limited. This study aimed to explore these interactions by using the SCN-soybean pathosystem as a model for assessing changes in soil fungal communities within agricultural soils infested with PPN. Our results suggest that SCN drives the selection of a specific group of fungi in field soil. While similar results have been seen with PPN infestation in roots and rhizosphere, this study is the first to demonstrate this effect for sedentary PPN in agricultural soil across different geographical locations (Elhady et al. 2021; Poll et al. 2007; Strom et al. 2020; Topalović and Heuer 2019).

Edaphoclimatic differences in study area and fungal community similarities. Regional scale variations in edaphoclimatic factors were observed across the sampling area. These differences could be determined only through the samples collected in 2021, during which edaphic factors were measured. Following PCA, samples that were geographically close and with similar edaphoclimatic profiles were used to delineate regions for our complete data set, including the 2019 samples. Using this information through a

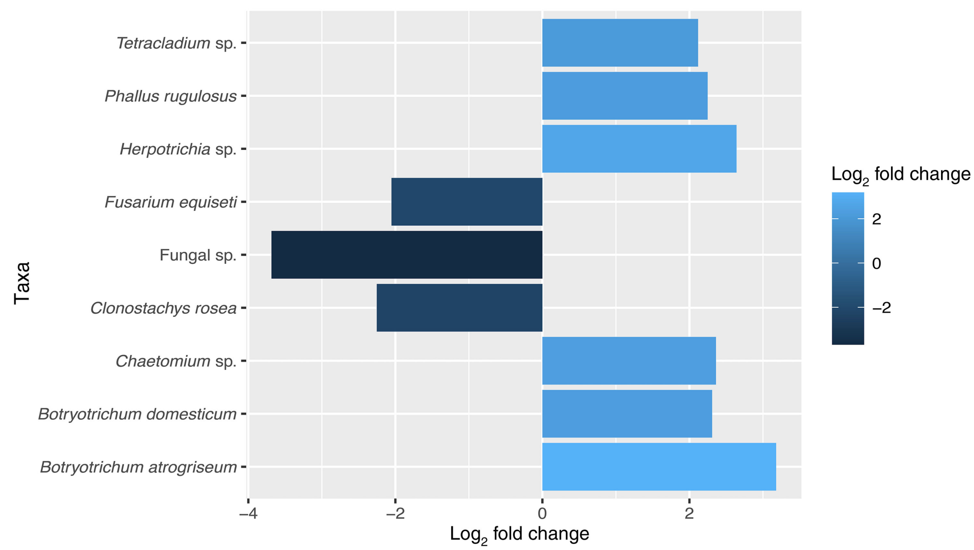


Fig. 6. Differential abundance of fungal taxa between samples with high soybean cyst nematode (SCN) abundance and samples with no SCN detected. Taxa with positive $\log_2$ fold change (light blue) were enriched in samples without SCN, whereas taxa with negative $\log_2$ fold change (dark blue) were enriched at high nematode abundance.

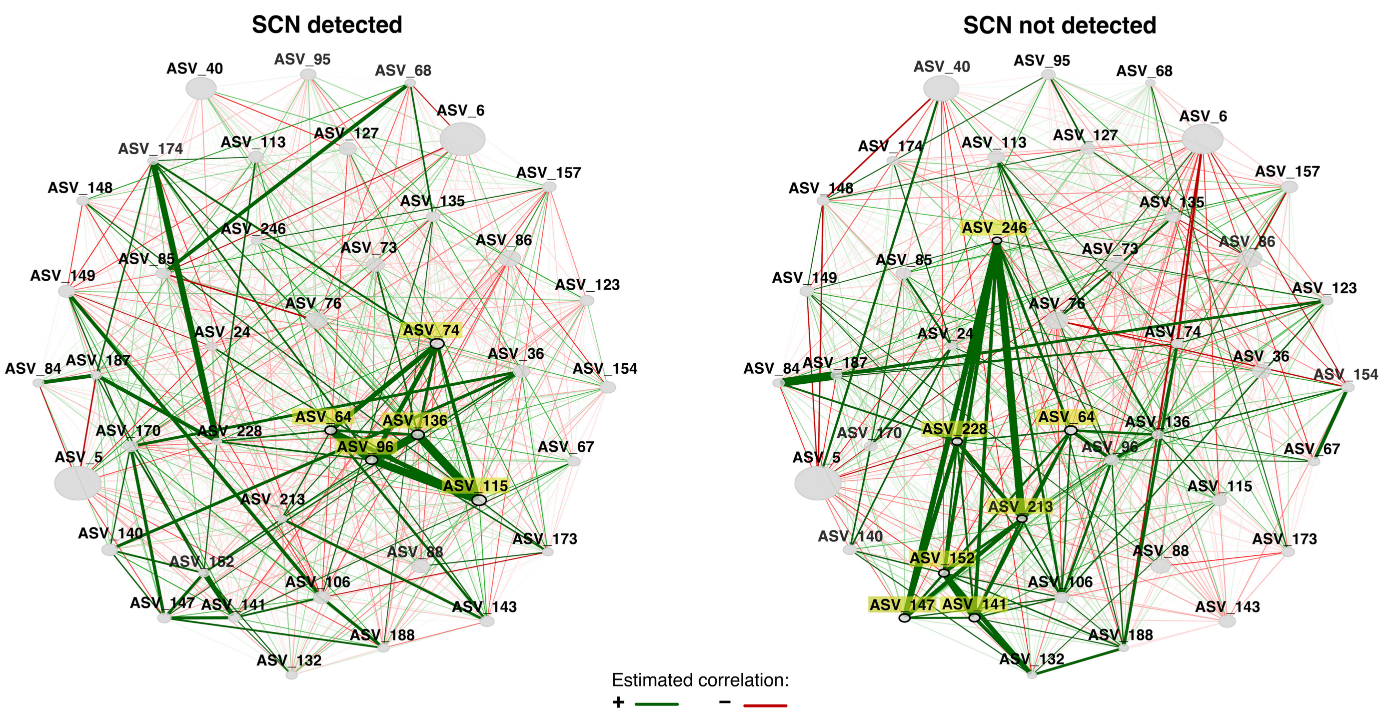


Fig. 7. Differential networks of significantly distinct fungal community correlations between samples with soybean cyst nematode (SCN) and without SCN detection ($P \leq 0.01$, correlation threshold ≥ 0.4). Each gray oval represents a fungal taxon, acting as nodes within the network. Node size reflects the relative abundance of the taxa in the community. Hub nodes (taxa) are highlighted in yellow. Green lines represent positive correlations between fungal community members, while red lines represent negative correlations. The weight of each line indicates the strength of the correlation.

K-near-neighbor analysis we demonstrated that the fungal communities of samples from the same region were significantly more similar to each other than those of samples from different regions (Fig. 3). This finding is in accordance with previous literature that demonstrates the significant impact that soil characteristics such as organic matter, pH, and nutrient status have on fungal community composition (Chen et al. 2021; Tedersoo et al. 2014). Additionally, previous studies have shown that MAT and MAP, which were also included in our analysis, significantly impact the composition of fungi in soil (Averill et al. 2019; Tedersoo et al. 2014). These factors were identified as significant contributors to the variation in regional edaphoclimatic conditions through PCA (Supplementary Fig. S2). Previous studies have shown legacy effects of cropping history on soil microbial communities associated to corn and soybean production (Benitez et al. 2017). However, in our study, a prior crop history effect was not detected, likely owing to the predominance of the corn-soybean rotation, though crop legacy may still hold biological significance in the studied areas.

SCN influences fungal community composition regardless of region. The soil fungal community composition was significantly influenced by the region of sample collection. Additionally, the abundance of SCN had a measurable effect on the soil fungi. Through indicator species analysis, several fungal ASV were identified as indicators of various levels of SCN abundance in the soil. Notably, at high levels of SCN abundance, a greater number of ASV were identified as indicators. In contrast, for samples with no detectable presence of SCN, only one ASV was identified as an indicator species. This suggests that SCN abundance can drive selection for a subset of fungi in soil. However, because regional effects were significant it was important to demonstrate that these indicator species were not the result of overlap between preferred edaphoclimatic conditions of the nematode with the indicator species. Samples with different levels of SCN abundance were collected from all regions. The samples displaying the smallest amount of variability in their indicator community composition were those with high levels of SCN abundance, regardless of the collection location (Fig. 4).

Differences in fungal communities at various SCN abundances. Core community analysis revealed specific fungi that were consistently present in most collected samples, at different levels of SCN abundance. Two putative soybean pathogenic fungi were part of the core community of SCN-infested soil. One of these was *F. virguliforme*, the causal agent of sudden death syndrome of soybean (SDS), also a core member in soil with high SCN abundance. Previous studies have found that *F. virguliforme* can have synergistic interactions with SCN, forming multipathogen disease complexes (Xing and Westphal 2013). SDS has been increasingly reported in the state of Ohio. During the 2023 growing season our team received samples infected with the pathogen from 15 Ohio counties. Here we found that these pathogens might be co-occurring frequently in Ohio soybean fields, which is concerning. As climatic conditions change, more areas in the state might be conducive to SDS development (Roth et al. 2020). This is also the case for *Corynespora cassiicola*, the causal agent of target spot of soybean, a disease that has not yet been reported in Ohio, but that was part of the core fungal community of low SCN abundance samples. This fungus has also been previously isolated from SCN cysts and eggs (Carris et al. 1989) and has been found to be highly abundant within soybean roots under SCN disease pressure (Strom et al. 2020). It is important to highlight that isolates of *C. cassiicola* will display a varied range of pathogenicity, with some not pathogenic to soybean (Dixon et al. 2009). Further studies could examine whether *C. cassiicola* commonly present in Ohio fields infested with low levels of SCN exhibit pathogenicity toward soybean. Because only the 18S-ITS region was utilized to identify the two taxa, it will be important to independently verify the taxonomy of these fungi using a different method to confirm their identity in further research. Nonetheless, these examples highlight potential interactions between nematode and fungal pathogens that infect the same plant host.

On the other hand, several members of the core community have previously been identified as potential nematode antagonists. These are members from *Mortierella*, *Exophiala*, *Fusarium*, and *Clonostachys* spp. Several species within these groups of fungi have been associated with SCN at different developmental stages and have shown nematophagous activity in previous studies (Haarith et al. 2020). For instance, *F. equiseti* was identified as a fungal core community member, recovered in both high and low levels of SCN. This fungus has been associated with decreased levels of SCN abundance in fields under long-term continuous soybean monoculture (Nitao et al. 2001; Song et al. 2017). On the other hand, the nematophagous fungus *Clonostachys rosea* was found to be prevalent in all soil samples. However, both nematode antagonists were found to be significantly enriched in samples with high SCN levels compared with those where SCN was not detected. Although it is not possible to determine whether these fungi are exhibiting antagonism toward SCN without further testing, this could potentially be the case. During the development of a disease, when antagonistic interactions occur, it is necessary to have a high and prolonged pressure of soil pathogens for an antagonist to increase in abundance and eventually reduce the pathogen pressure (Schlatter et al. 2017). These findings could potentially explain the distribution of nematode antagonists identified in this study.

Fungal community network differences associated with SCN detection. To further understand how fungal communities differed between soil samples with and without SCN, a differential community network analysis was performed. When compared, these networks shared only one hub taxon. All hub taxa of the SCN-detected network belong to the Mortierellaceae family, pointing to an important role of this group within fungal networks under SCN presence. For instance, in the past, *Mortierella* sp. has been identified as both potential nematode antagonists and plant growth promoters that aid plants to cope with pathogen infection (Johnson et al. 2019; Ozimek et al. 2018; Strom et al. 2020). They also have dual roles as plant endophytes and saprophytes. Therefore, there are several possible explanations for their role as hubs in this network. They might be more interconnected to the network in the presence of the nematode owing to nutrient mobilization to the rest of the fungal community through nematode consumption. Another possibility is that this group of fungi is being actively recruited by the plant during nematode infection for their plant growth-promoting activity through increased nutrient turnover in the soil, which in turn would also make more nutrients available to other members of the fungal network. In contrast, hub taxa belonging to several different fungal families were seen in samples with no SCN, indicating that the fungal network is not as reliant on a specific taxonomic group in the absence of SCN. However, within this network, ASV_246 appeared to play a crucial role in connecting all other hub taxa. Unfortunately, this hub could not be taxonomically identified beyond kingdom level.

Core community members *C. rosea* (ASV_5) and *F. equisteti* (ASV_6), displayed a greater number of negative correlations with other fungal community members in the absence of SCN detection in the soil. This is consistent with the observations of the differential abundance analysis, which showed that both fungi had higher abundance when SCN was present compared with when no SCN was detected. This indicates that the two fungi are promising candidates for further testing as SCN antagonists. In all, the differential network analysis demonstrated how the shared fungi in the sampled soils associate significantly differently when SCN is present.

Conclusions. In this study, we demonstrated the impact of SCN on the soil fungal community. We have shown that SCN can drive the selection of specific groups of fungi, despite the significant influence of edaphoclimatic factors and the potential effect of current or previous crops. Additionally, our research revealed variations in fungal community associations in soils when nematodes are present. Further studies on these variations can assist in investigating the development of multiple pathogen interactions between plant pathogenic fungi and SCN. Our results also identified fungi that should be further studied for their potential as nematode antagonists especially in Ohio, where sampling took place. This could result in additional biocontrol solutions for growers. In the future, additional studies should be conducted to better understand the mechanisms driving the differences described here. Are these differences solely influenced by the plant's response to nematode infection or do they involve interactions between the nematode and surrounding fungal community as well? Answering some of these questions can aid in identifying management practices that encourage the assembly of fungal communities that help reduce SCN disease pressure.

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